

Kronecker - Capelli

System is compatible if $\text{rank}(A) = \text{rank}(\bar{A})$.

Rouché

system is compatible if all characteristic minors are zero.

Cramer

a system has a unique solution

$$x_i = \frac{\Delta x_i}{\Delta}$$

Gauss

getting $\bar{A}$ in its echelon form (zero underneath the main diag.)

Gauss - Jordan

getting $\bar{A}$ to an identity form (ones on the main diag.)

Use only operations on rows! (zeros otherwise)

5) a)
$$\begin{cases} 2x + 2y + 3z = 3 \\ x - y = 1 \\ -x + 2y + z = 2 \end{cases}$$

$$\Rightarrow \bar{A} = \left(\begin{array}{ccc|c} 2 & 2 & 3 & 3 \\ 1 & -1 & 0 & 1 \\ -1 & 2 & 0 & 2 \end{array} \right) \xrightarrow{R_2 \leftrightarrow R_1} \left(\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 2 & 2 & 3 & 3 \\ -1 & 2 & 0 & 2 \end{array} \right) \begin{array}{l} R_2 \leftarrow R_2 - 2R_1 \\ R_3 \leftarrow R_3 + R_1 \end{array}$$

$$\sim \left(\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 4 & 3 & 1 \\ 0 & 1 & 1 & 3 \end{array} \right) \xrightarrow{R_3 \leftrightarrow R_2} \left(\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 4 & 3 & 1 \end{array} \right) \xrightarrow{R_3 \leftarrow R_3 - 4R_2}$$

$$\sim \left(\begin{array}{ccc|c} 1 & -1 & 0 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & -1 & -11 \end{array} \right) \Rightarrow \begin{cases} x - y = 1 \\ y + z = 3 \\ -z = -11 \end{cases}$$

Gauss.
(echelon form; Gauss steps.)

$$\begin{aligned} z &= 11 \\ y &= 3 - 11 = -8 \\ x &= 1 + y = -7 \end{aligned}$$

if $0 = -11$
 $\Rightarrow$ incompatible.

$$\begin{pmatrix} 1 & -1 & 0 & 1 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & -1 & -11 \end{pmatrix} \xrightarrow[R_1 = R_1 + R_2]{R_3 = -R_3} \begin{pmatrix} 1 & 0 & 1 & 4 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & +11 \end{pmatrix} \xrightarrow[R_2 = R_2 - R_3]{R_1 = R_1 - R_3} \sim \begin{pmatrix} 1 & 0 & 0 & -7 \\ 0 & 1 & 0 & -8 \\ 0 & 0 & 1 & +11 \end{pmatrix} \Rightarrow \begin{cases} x = -7 \\ y = -8 \\ z = 11 \end{cases}$$

Gauss-Jordan

① $A = \begin{pmatrix} 1 & 4 & 2 \\ 2 & 3 & 1 \\ 3 & 0 & -1 \end{pmatrix}$

$$X = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, B = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

a) Show that A invertible. $A^{-1} = ?$

b) Solve $AX = B$.

a) $\det A = \begin{vmatrix} 1 & 4 & 2 \\ 2 & 3 & 1 \\ 3 & 0 & -1 \end{vmatrix} = -3 + 12 + 8 - 18 = 9 - 10 = -1 \neq 0$
 $\Rightarrow A$ invertible.

$$A^t = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 3 & 0 \\ 2 & 1 & -1 \end{pmatrix}$$

$$a_{1,1}^* = (-1)^{1+1} \cdot \begin{vmatrix} 3 & 0 \\ 1 & -1 \end{vmatrix} = 1 \cdot (-3) = -3.$$

$$a_{1,2}^* = (-1)^{1+2} \cdot \begin{vmatrix} 4 & 0 \\ 2 & -1 \end{vmatrix} = -(-4) = 4.$$

$$a_{1,3}^* = (-1)^{1+3} \cdot \begin{vmatrix} 4 & 3 \\ 2 & 1 \end{vmatrix} = 4 - 6 = -2.$$

$$a_{2,1}^* = (-1)^{2+1} \cdot \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} = -(-2 - 3) = 5.$$

$$a_{2,2}^* = (-1)^{2+2} \cdot \begin{vmatrix} 1 & 3 \\ 2 & -1 \end{vmatrix} = -1 - 6 = -7.$$

$$a_{3,1}^* = (-1)^{3+1} \cdot \begin{vmatrix} 2 & 3 \\ 3 & 0 \end{vmatrix} = -9$$

$$a_{3,3}^* = (-1)^{3+3} \cdot \begin{vmatrix} 1 & 2 \\ 4 & 3 \end{vmatrix} = 3 - 8 = -5.$$

$$a_{3,2}^* = (-1)^{3+2} \cdot \begin{vmatrix} 1 & 3 \\ 4 & 0 \end{vmatrix} = -(-12) = 12.$$

$$\begin{matrix} a_{2,3}^* \\ 2(-1)^{5+2} \cdot \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} \\ 2(-1)^{5+3} \cdot \begin{vmatrix} 1 & 4 \\ 2 & 3 \end{vmatrix} \end{matrix}$$

$$\Rightarrow A^* = \begin{pmatrix} -3 & 4 & -2 \\ 5 & -7 & 3 \\ -9 & 12 & -5 \end{pmatrix}$$

$$A^{-1} = \frac{1}{\det A} \cdot A^* = -1 \cdot \begin{pmatrix} -3 & 4 & -2 \\ 5 & -7 & 3 \\ -9 & 12 & -5 \end{pmatrix} = \begin{pmatrix} 3 & -4 & 2 \\ -5 & 7 & -3 \\ 9 & -12 & 5 \end{pmatrix}$$

$$A^{-1} \cdot AX = B \Rightarrow X = A^{-1} \cdot B$$

$$= \begin{pmatrix} 3 & -4 & 2 \\ -5 & 7 & -3 \\ 9 & -12 & 5 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 0 \\ 2 \end{pmatrix}$$

$$= \begin{pmatrix} 0 \cdot 3 - 4 \cdot 0 + 2 \cdot 2 \\ 0 \cdot (-5) + 7 \cdot 0 - 3 \cdot 2 \\ 0 \cdot 9 - 12 \cdot 0 + 5 \cdot 2 \end{pmatrix} = \begin{pmatrix} 4 \\ -6 \\ 10 \end{pmatrix} \Rightarrow \begin{cases} x = 4 \\ y = -11 \\ z = 19 \end{cases}$$

$$= \begin{pmatrix} 0 \\ 1 \\ -2 \end{pmatrix}$$

$$\textcircled{2} \text{ a) } \begin{cases} x_1 + x_2 + x_3 - 2x_4 = 5 \\ 2x_1 + x_2 - 2x_3 + x_4 = 1 \\ 2x_1 - 3x_2 + x_3 + 2x_4 = 3 \end{cases}$$

Kronecker-Capelli: $\text{rank}(A) = \text{rank}(\bar{A}) \Rightarrow (S) \text{ compatible}$

$$\bar{A} = \left(\begin{array}{cccc|c} 1 & 1 & 1 & -2 & 5 \\ 2 & 1 & -2 & 1 & 1 \\ 2 & -3 & 1 & 2 & 3 \end{array} \right)$$

Not Gauss!

$$\det \dots = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 1 & -2 \\ 2 & -3 & 1 \end{vmatrix} = 1 - 6 - 4 - 2 + 6 - 2 = -7 \neq 0$$

$$\Rightarrow \text{rank}(A) = 3 = \text{rank}(\bar{A}) \Rightarrow (S) \text{ comp.}$$

(we cannot make rank 4)

$$\Rightarrow \begin{cases} x_1 + x_2 + x_3 = 5 + 2x_4 & (1) \\ 2x_1 + x_2 - 2x_3 = 1 - x_4 & (2) \\ 2x_1 - 3x_2 + x_3 = 3 - 2x_4 & (3) \end{cases}$$

$$(2) - (3) \Rightarrow \begin{cases} -2x_2 - 3x_3 = -2 + 3x_4 \\ x_1 + x_2 + x_3 = 5 + 2x_4 \cdot 1.2 \end{cases} \Rightarrow \begin{cases} -2x_2 - 3x_3 = -2 + 3x_4 \\ 2x_1 + 2x_2 + 2x_3 = 10 + 4x_4 \end{cases}$$

$$\Rightarrow 2x_1 - x_3 = 8 + 7x_4 \Rightarrow x_3 = 2x_1 - 8 - 7x_4$$

$$\hookrightarrow (3): 2x_1 - 3x_2 + 2x_1 - 8 - 7x_4 = 3 - 2x_4 \Rightarrow 4x_1 - 3x_2 - 7x_4 - 8 = 3 - 2x_4$$

~~$\Rightarrow x_2$~~

$$\begin{cases} x_1 + x_2 + x_3 = 5 + 2x_4 & (1) \\ 2x_1 + x_2 - 2x_3 = 1 - x_4 & (2) \\ 2x_1 - 3x_2 + x_3 = 3 - 2x_4 & (3) \end{cases} \quad (-)$$

$$\begin{cases} 4x_2 - 3x_3 = -2 + x_4 \\ x_1 + x_2 + x_3 = 5 + 2x_4 \quad | \cdot 4 \end{cases} \quad \Rightarrow 4x_2 = -2 + x_4 + 3x_3$$

$$\Rightarrow \begin{cases} 4x_2 - 3x_3 = -2 + x_4 \\ 4x_1 + 4x_2 + 4x_3 = 20 + 8x_4 \end{cases} \quad \Rightarrow 4x_2 = -2 + x_4 + 3x_3$$

$$\Rightarrow x_2 = \frac{-2 + x_4 + 3x_3}{4}$$

$$4x_1 - 7x_3 = -22 - 7x_4$$

... $x_1 = 5 + 2x_4$

$$\Rightarrow 4(5 + 2x_4) - 7x_3 = -22 - 7x_4$$

$$\Rightarrow 20 + 8x_4 - 7x_3 = -22 - 7x_4$$

$$\Rightarrow 7x_3 = 44 + 15x_4$$

$$x_3 = \frac{44 + 15x_4}{7}$$

$$x_2 = \frac{-2 + x_4 + 3 \cdot \frac{44 + 15x_4}{7}}{4}$$

$$\begin{array}{r} 10 \\ 132 - \\ 14 \\ \hline 118 \end{array}$$

$$= \frac{-14 + 7x_4 + 132 + 45x_4}{28}$$

$$= \frac{52x_4 + 118}{28}$$

$$\begin{cases} x_1 = 2 \\ x_2 = x_4 + 1 \\ x_3 = x_4 + 2 \\ x_4 \end{cases}$$

$$(1) + (3) \Rightarrow \begin{cases} 5x_1 - 2x_2 + 2x_3 = 8 \\ 2x_1 + x_2 - 2x_3 = 1 - x_4 \end{cases} \quad (-)$$

$$\Rightarrow 5x_1 - x_2 = 9 - x_4 \Rightarrow x_2 = 5x_1 - 9 + x_4$$

$$8x_1 - 2(5x_1 - 9 + x_4) + 2x_3 = 8$$

$$\Rightarrow -7x_1 + 18 - 2x_4 + 2x_3 = 8$$

$\Rightarrow$

$$\textcircled{b) } \begin{cases} 2x + 5y + z = 7 \\ x + 2y - z = 3 \\ x + y - 4z = 2 \end{cases}$$

$$\bar{A} = \left(\begin{array}{ccc|c} 2 & 5 & 1 & 7 \\ 1 & 2 & -1 & 3 \\ \textcircled{1} & 1 & -4 & 2 \end{array} \right) \xrightarrow{L_2+L_3} \left(\begin{array}{ccc|c} 2 & 5 & 1 & 7 \\ 3 & 7 & 0 & 5 \\ 1 & 1 & -4 & 2 \end{array} \right)$$

$$\xrightarrow{L_3-L_2 \rightarrow L_3} \left(\begin{array}{ccc|c} 2 & 5 & 1 & 7 \\ 1 & 2 & -1 & 3 \\ 0 & -1 & -3 & -1 \end{array} \right) \xrightarrow{L_1-L_2 \rightarrow L_1} \left(\begin{array}{ccc|c} 1 & 3 & 2 & 4 \\ \textcircled{1} & 2 & -1 & 3 \\ 0 & -1 & -3 & -1 \end{array} \right) \xrightarrow{L_2-L_1 \rightarrow L_2} \left(\begin{array}{ccc|c} 1 & 3 & 2 & 4 \\ 0 & -1 & -3 & -1 \\ 0 & -1 & -3 & -1 \end{array} \right)$$

$$= \left(\begin{array}{ccc|c} 1 & 3 & 2 & 4 \\ 0 & -1 & -3 & -1 \\ 0 & \textcircled{-1} & -3 & -1 \end{array} \right) = \left(\begin{array}{ccc|c} 1 & 2 & -1 & 3 \\ 0 & 1 & 3 & 1 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$\Rightarrow \begin{cases} x + 2y - z = 3 \\ y + 3z = 1 \end{cases} \quad \begin{matrix} x = \alpha \\ \begin{cases} 2y - z = 3 - \alpha \\ y + 3z = 1 \Rightarrow y = 1 - 3z \end{cases} \end{matrix}$$

$$\Rightarrow 2(1 - 3z) - z = 3 - \alpha \Rightarrow 2 - 6z - z = 3 - \alpha \Rightarrow$$

$$\Rightarrow -7z = 3 - \alpha - 2 \Rightarrow z = \frac{1 - \alpha}{-7} = \frac{\alpha - 1}{7}$$

$$y = 1 - 3 \cdot \frac{\alpha - 1}{7} = \frac{7 - 3\alpha + 3}{7} = \frac{10 - 3\alpha}{7} \quad \frac{4 + 3\alpha}{7}$$

$$\Rightarrow S = \left\{ \left(x, \frac{4 + 3x}{7}, \frac{x - 1}{7} \right) \right\}^{x \in \mathbb{R}}$$

$$\textcircled{b) } \begin{cases} 2x_1 + x_2 + x_3 + x_4 = 1 \\ x_1 + 2x_2 - x_3 + 4x_4 = 2 \\ x_1 + 5x_2 - 4x_3 + 11x_4 = \lambda, \quad \lambda \in \mathbb{R} \end{cases}$$

$$\left(\begin{array}{cccc|c} 2 & 1 & 1 & 1 & 1 \\ 1 & 2 & -1 & 4 & 2 \\ 1 & 5 & -4 & 11 & \lambda \end{array} \right), \quad \begin{array}{c} \cancel{2} \quad \cancel{1} \quad \cancel{1} \\ 1 \quad 2 \quad -1 \\ 1 \quad 5 \quad -4 \end{array}$$

$$R_3 = 3R_2 - R_1$$

$$\left(\begin{array}{cccc|c} 2 & 1 & 1 & 1 & 1 \\ 1 & 2 & -1 & 4 & 2 \\ 0 & 0 & 0 & 0 & \lambda - 5 \end{array} \right)$$

if $\lambda - 5 \neq 0 \Rightarrow$ incomp.

$$\text{else } \lambda = 5 \Rightarrow \left(\begin{array}{cccc|c} 2 & 1 & 1 & 1 & 1 \\ 1 & 2 & -1 & 4 & 2 \end{array} \right)$$

$$\Rightarrow \begin{cases} 2\alpha + \beta + x_3 + x_4 = 1 \\ \alpha + 2\beta - x_3 + 4x_4 = 2 \end{cases} \quad (+)$$

$$3\alpha + 3\beta + 5x_4 = 3 \\ \Rightarrow x_4 = \frac{3 - 3\alpha - 3\beta}{5}$$

$$x_3 = 1 - 2\alpha - \beta - x_4 \\ = 1 - 2\alpha - \beta - \frac{3 - 3\alpha - 3\beta}{5}$$

$$\begin{matrix} \alpha = x_1 \\ \Rightarrow \beta = x_2 \end{matrix}$$